Structural Aliasing Analysis in Time Series Forecasting using Chronos

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Executive Summary

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During the preparation of this work, the authors used AI-based tools, including GPT-5.5, GPT-5.4, Gemini 3.1 Pro, and Sonnet 4.6, to support language refinement and improve clarity. All generated or suggested content was carefully reviewed, revised where necessary, and validated by the authors, who assume full responsibility for the final content of this work.

Introduction

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$$\cos^3\theta = \frac{1}{4}\cos\theta + \frac{3}{4}\cos 3\theta \tag{1}$$

Automatically referencing an equation number using its label: Equation 1.

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1 Formal Definition of Patch-Based tokenization

1.1 Setup and Formal Definition of the Tokenisation Operator

To rigorously understand the failure modes of patch-based architectures like Chronos-Bolt, we must first formally define the tokenisation process. Let $x[n]$ represent a discrete-time, univariate signal where the index $n \in \mathbb{Z}$ is sampled at a continuous frequency of f_s.

Unlike classical autoregressive models that process temporal data one scalar time-step at a time, modern foundation models employ *patch-based tokenisation*. This technique groups contiguous samples to radically reduce the effective sequence length and improve the computational efficiency of the transformer's attention mechanisms. This is achieved by sliding a window of fixed size P (the *patch length*) across the signal, moving forward by a step size of S (the *stride*).

We can extract this sequence of tokens as follows:

$$\mathbf{p}_k = \big(x[kS],\ x[kS+1],\ \ldots,\ x[kS+P-1]\big) \in \mathbb{R}^P, \qquad k \in \mathbb{N}. \tag{2}$$

Example 1 (Constructing patches by hand)**.** *Let $x = [x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7]$ with $P = 4$ and $S = 2$. The first three patches are:*

$$\mathbf{p}_0 = [x_0, x_1, x_2, x_3], \qquad \mathbf{p}_1 = [x_2, x_3, x_4, x_5], \qquad \mathbf{p}_2 = [x_4, x_5, x_6, x_7].$$

Because $S < P$, consecutive patches overlap by $P - S = 2$ samples (e.g. $\mathbf{p}_0$ and $\mathbf{p}_1$ share x_2 and x_3). With $S = P = 4$ (no overlap), the patches would instead be $\mathbf{p}_0 = [x_0, \ldots, x_3]$, $\mathbf{p}_1 = [x_4, \ldots, x_7]$, with no shared samples at all.

Definition 1 (Patch Tokenisation Operator)**.** *We can formalise this extraction as a linear map acting on the bounded signal space. The map $\mathscr{T} : \ell^\infty(\mathbb{Z}) \to (\mathbb{R}^P)^{\mathbb{N}}$, as defined above, is referred to as the* patch tokenisation operator. *When collecting an arbitrary number of K patches from a signal vector $\mathbf{x}$, the operation can be represented as a matrix multiplication:*

$$\mathbf{P} = \mathscr{T}\{x\} = \mathbf{W} \cdot \mathbf{x}. \tag{3}$$

Here, $\mathbf{W} \in \{0,1\}^{KP \times N}$ is a block-binary selection matrix. *Its k-th block consists of P rows that act as a mask, picking out exactly P consecutive signal samples starting at the temporal index kS.*

Example 2 (Explicit matrix form)**.** *Take $N = 6$, $P = 3$, $S = 3$ (non-overlapping), giving $K = 2$ patches. The selection matrix and the resulting patch matrix are:*

$$\mathbf{W} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \mathbf{P} = \mathbf{W}\mathbf{x} = \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix},$$

which, when reshaped into $K \times P = 2 \times 3$ block form, gives $\mathbf{p}_0 = [x_0, x_1, x_2]$ and $\mathbf{p}_1 = [x_3, x_4, x_5]$. Each row of $\mathbf{W}$ contains exactly one non-zero entry (a 1) at the position of the corresponding signal sample, confirming the pure selection nature of the operator.

The operator $\mathscr{T}$ is *linear* because:

1. $\mathscr{T}\{ax\} = a\mathscr{T}\{x\}$ for any scalar a — scaling the signal scales every patch proportionally.
2. $\mathscr{T}\{x + y\} = \mathscr{T}\{x\} + \mathscr{T}\{y\}$ — the patch of a sum equals the sum of the patches.

Linearity is the key property that makes the rank-collapse analysis in Section 1 tractable.

1.2 The Mechanics of Phase-Locking and Degeneracy

The fundamental vulnerability of the patch tokenisation operator emerges when the periodic structure of the input signal aligns perfectly with the model's fixed stride grid. We refer to this specific synchronisation as *phase-locking*.

Proposition 1 (Condition for Token Identity)**.** *Two consecutive patches, $\mathbf{p}_k$ and $\mathbf{p}_{k+1}$, will be mathematically identical if and only if the signal amplitude remains constant across the stride shift for every element within the observation window:*

$$x[n] = x[n+S] \quad \forall\, n \in \{kS,\, kS+1,\, \dots,\, kS+P-1\}. \tag{4}$$

Globally, if the underlying continuous signal is a pure sinusoid (or any periodic signal) with a fundamental period of T_0 samples, this exact alignment occurs whenever the stride S spans an integer multiple of the signal's period:

$$\boxed{S = m \cdot T_0, \quad m \in \mathbb{Z}^+.} \tag{5}$$

Substituting $T_0 = f_s/f$, we can express this condition directly in terms of the signal's continuous frequency f and the sampling frequency f_s:

$$f = m \cdot \frac{f_s}{S}, \qquad m \in \mathbb{Z}^+. \tag{6}$$

Proof. To satisfy $\mathbf{p}_k = \mathbf{p}_{k+1}$ strictly component-wise, it is required that $x[kS+j] = x[(k+1)S+j]$ for every internal patch index $j \in \{0, \dots, P-1\}$. For a periodic signal with $T_0 = f_s/f$, this strict equality holds whenever the stride shift S represents a traversal of exactly m full wave cycles. In such instances, the sliding window captures the geometrically identical segment of the waveform at every step, and the proposition follows. □

Example 3 (Phase-locking with a discrete sinusoid)**.** *Let $f_s = 8\,Hz$, $f = 2\,Hz$, so $T_0 = f_s/f = 4$ samples. Choose $S = 4$ ($= 1 \cdot T_0$) and $P = 4$. The signal values are:*

n	*0*	*1*	*2*	*3*	*4*	*5*	*6*	*7*
$x[n]$	0	1	0	−1	0	1	0	−1

*Then $\mathbf{p}_0 = [0, 1, 0, -1]$ and $\mathbf{p}_1 = [0, 1, 0, -1]$: the two patches are **identical**. If instead we set $S = 8 = 2T_0$, every patch still starts at phase 0, and the result is the same. Conversely, with $S = 3$ (not a multiple of $T_0 = 4$), we obtain $\mathbf{p}_0 = [0, 1, 0, -1]$ and $\mathbf{p}_1 = [-1, 0, 1, 0]$, which are distinct — no phase-locking occurs.*

1.3 The Consequence: Loss of Observability

When the phase-locking condition $f = m \cdot (f_s/S)$ is met, we achieve a state where $\mathbf{p}_k = \mathbf{p}_{k+1}$ for all k. The token matrix $\mathbf{P}$ undergoes a mathematical *rank collapse*, reducing its rank to exactly 1.

This creates a *degenerate token sequence*, which induces severe representational failures within the network:

- **Complete Temporal Blindness.** Because the transformer processes a sequence of completely identical vectors, the temporal variation of the actual underlying signal becomes entirely invisible to the model's self-attention mechanisms. The model cannot perceive that a wave is oscillating; it simply perceives a constant, flat embedding.
- **Observational Equivalence and Aliasing.** Any two distinct signals that differ only by a phase shift of S samples produce the exact same sequence of tokens. They become observationally indistinguishable to the decoder, making it impossible for the model to correctly reconstruct or forecast the true phase of the original signal.

Example 4 (Two signals, one token sequence). *With $f_s = 8\,Hz$, $f = 2\,Hz$, $S = 4$, consider two phase-shifted signals:*

$$x_1[n] = \cos\!\left(\tfrac{2\pi\cdot 2\,n}{8}\right), \qquad x_2[n] = \cos\!\left(\tfrac{2\pi\cdot 2\,n}{8} + \tfrac{\pi}{2}\right).$$

x_1 has values $[1, 0, -1, 0, 1, 0, -1, 0, \ldots]$ and x_2 has values $[0, -1, 0, 1, 0, -1, 0, 1, \ldots]$. Their patch sequences with $P = 4$:

$$\begin{aligned} x_1 : \mathbf{p}_0 &= [1, 0, -1, 0], \quad \mathbf{p}_1 = [1, 0, -1, 0], \quad \ldots \\ x_2 : \mathbf{p}_0 &= [0, -1, 0, 1], \quad \mathbf{p}_1 = [0, -1, 0, 1], \quad \ldots \end{aligned}$$

Each signal individually produces a constant *token sequence. Although the two constant sequences differ from each other here, the decoder cannot infer the phase trajectory over time from either one — both look as if the signal is not evolving.*

- **Soft Degradation at Boundary Frequencies.** This failure is not isolated to exact integer multiples. Even when patches are not perfectly identical but merely highly correlated (i.e. when $S \approx mT_0$), the sequence suffers a *soft* loss of observability. The effective information bandwidth is severely restricted, explaining why predictive accuracy degrades smoothly around these critical frequency boundaries before fully collapsing at the exact harmonic frequencies.

For example, in the specific architecture of Chronos-Bolt, the sampling frequency is $f_s = 512\,\text{Hz}$ and the stride is $S = 16$. According to our derived condition $f = m \cdot (512/16)$, the fundamental patch frequency is precisely $32\,\text{Hz}$. Consequently, any input signal at $32\,\text{Hz}$, $64\,\text{Hz}$, $96\,\text{Hz}$, or other integer harmonics will trigger this structural aliasing, resulting in an immediate and sharp collapse in the model's predictive accuracy.

2 Phase-locking and Degeneration Analysis

2.1 The Patch-Induced Frequency

The tokeniser introduces a characteristic *patch frequency*:

$$f_{\text{patch}} = \frac{f_s}{S}. \tag{7}$$

This is the rate at which new patches are produced (in Hz, referred to original time), and it plays a role analogous to a resampling rate in the token domain. Intuitively, f_{patch} sets the *temporal resolution* of the token sequence: a smaller stride S yields a higher patch frequency and thus a finer-grained view of the signal's evolution, while a larger stride coarsens the view and reduces computational cost.

Example 5 (Computing f_{patch} for Chronos-Bolt)**.** *Chronos-Bolt uses $f_s = 512\,Hz$ and stride $S = 16$. Therefore:*

$$f_{\text{patch}} = \frac{512}{16} = 32\,Hz.$$

This means the model receives one new token every $1/32 \approx 31.25\,ms$, regardless of how fast the underlying signal actually varies. Any signal whose periodicity is commensurate with this $32\,Hz$ grid is at risk of phase-locking, as formalised below.

2.2 Phase-Locking Condition

Consider a sinusoidal signal $x[n] = A\cos\big(2\pi f\, n/f_s + \phi\big)$.

Definition 2 (Phase-Locking)**.** *The signal* phase-locks *to the patch grid when its period $T_0 = f_s/f$ (in samples) is* commensurate *with S:*

$$\frac{S \cdot f}{f_s} \in \mathbb{Q}. \tag{8}$$

The degenerate *(critical) case occurs when this ratio is an integer:*

$$f = m \cdot \frac{f_s}{S} = m \cdot f_{\text{patch}}, \qquad m \in \mathbb{Z}^+, \tag{9}$$

i.e. f is a ***harmonic of*** f_{patch}.

The three regimes — integer ratio, rational ratio, and irrational ratio — produce qualitatively different behaviours, summarised below.

Example 6 (Three regimes of phase-locking)**.** *Let $f_s = 100\,Hz$ and $S = 10$, so $f_{\text{patch}} = 10\,Hz$.*

Case 1 — Integer ratio (full degeneracy). *$f = 20\,Hz$, so $T_0 = 5$ samples and $S/T_0 = 2 \in \mathbb{Z}$. Every patch starts at exactly the same phase of the sinusoid; all patches are identical and the token sequence is constant.*

Case 2 — Rational ratio (partial degeneracy). *$f = 15\,Hz$, so $T_0 = 100/15 \approx 6.67$ samples and $S/T_0 = 3/2 \in \mathbb{Q} \setminus \mathbb{Z}$. Patches alternate between two distinct phase offsets: the sequence is periodic with period 2, not constant, so some information survives.*

Case 3 — Irrational ratio (no phase-locking). *$f = 10\sqrt{2} \approx 14.14\,Hz$, so $S \cdot f/f_s = \sqrt{2} \notin \mathbb{Q}$. The starting phase advances by an irrational fraction of 2π at each step; the patches never repeat and the full phase information is preserved.*

2.3 Effect of Patch Size P

A single patch of P samples spans a duration P/f_s seconds and covers $P \cdot f / f_s$ cycles of the signal. When

$$\frac{P \cdot f}{f_s} \in \mathbb{Z}^+, \quad \text{i.e.} \quad f = k \cdot \frac{f_s}{P}, \quad k \in \mathbb{Z}^+, \tag{10}$$

the patch contains an exact integer number of cycles: its content is *internally redundant*.

Example 7 (Within-patch redundancy). *Let $P = 8$, $f_s = 8\,Hz$, $f = 2\,Hz$ ($T_0 = 4$ samples). A single patch contains $P \cdot f / f_s = 8 \cdot 2/8 = 2$ complete cycles:*

$$\mathbf{p} = [0, 1, 0, -1,\ 0, 1, 0, -1].$$

The second half is an exact copy of the first half. The patch is internally redundant: it carries no more information than a patch of length $P/2 = 4$ would. Any linear feature extractor applied to this patch can at best recover the waveform shape, but cannot distinguish whether the signal has been observed for 4 or 8 samples.

The most severe degeneracy occurs when *both* the stride condition and the patch-size condition hold simultaneously:

$$f = \frac{\ell \cdot f_s}{\mathrm{lcm}(P, S)}, \qquad \ell \in \mathbb{Z}^+. \tag{11}$$

At these frequencies, each patch is internally redundant *and* all patches are identical — the token sequence is maximally uninformative.

Example 8 (Maximal degeneracy). *Let $f_s = 100\,Hz$, $P = 6$, $S = 4$, so $\mathrm{lcm}(6, 4) = 12$. The joint condition requires $f = \ell \cdot 100/12 \approx 8.33\,\ell\,Hz$. For $\ell = 1$: $f \approx 8.33\,Hz$, $T_0 = 12$ samples. Every patch spans exactly $6/12 = 0.5$ cycles (internally redundant in the sense that T_0 divides $\mathrm{lcm}(P,S)$), and the stride advances by exactly $4/12 = 1/3$ of a period, which is rational — triggering the partial phase-lock of Case 2 above and producing a two-periodic token sequence with minimal diversity.*

2.4 Token-Space Degeneracies

At $f = m f_s/S$ (harmonics of f_{patch}).

- All patches are translates of the same waveform by a phase that is a multiple of 2π: they become **identical** (cf. Point 1).
- The token sequence is constant; it encodes **no phase information**.
- Two signals at the same frequency but different initial phases (shifted by any multiple of S samples) map to the same token sequence: *token-space invariance*.

At $f = k f_s/P$ (but not harmonics of f_{patch}).

- Each patch is internally periodic, so its content is *within-patch redundant*.
- Inter-patch variation still exists (partial degeneracy): the model retains some discriminative power, but each token is a compressed, information-poor representation of the signal at that time step.

2.5 Overlap, Stride, and Blind-Spot Frequencies

When $S < P$ (overlapping patches), consecutive patches share $P - S$ samples, partially mitigating the aliasing effect even near harmonic frequencies. Intuitively, the shared samples act as a *bridge* between consecutive tokens, preserving some phase continuity even when the stride would otherwise trigger phase-locking.

Example 9 (How overlap rescues phase information). *Let $f_s = 100\,Hz$, $f = 25\,Hz$ ($T_0 = 4$ samples), $P = 6$, $S = 4$ ($= T_0$, so the stride condition is met). Without overlap ($S = P = 4$): $\mathbf{p}_0 = [0, 1, 0, -1]$ and $\mathbf{p}_1 = [0, 1, 0, -1]$ — identical. With overlap ($S = 4 < P = 6$):*

$$\mathbf{p}_0 = [x_0, x_1, x_2, x_3, x_4, x_5] = [0, 1, 0, -1, 0, 1],$$

$$\mathbf{p}_1 = [x_4, x_5, x_6, x_7, x_8, x_9] = [0, 1, 0, -1, 0, 1].$$

In this particular case the patches are still identical because $T_0 = 4$ also divides $P = 6$ (partially) — but for a generic P not divisible by T_0, the shared samples would introduce a phase offset and break the degeneracy. This illustrates that overlap is a necessary, but not always sufficient, safeguard.

For $S \geq P$ (no overlap or gaps), the aliasing is maximal; the **blind-spot frequencies** are:

$$\boxed{f_{\text{blind}} = m \cdot \frac{f_s}{S}, \qquad m = 1, 2, 3, \ldots, \left\lfloor \frac{S}{2} \right\rfloor.} \tag{12}$$

The upper limit follows from the Nyquist criterion applied to the token sequence, whose effective sample rate is f_s/S. These consequences follow directly from what is expressed in [**paganiPreliminaryInsightsChronos2026b**].

Example 10 (Blind-spot table for Chronos-Bolt). *With $f_s = 512\,Hz$ and $S = 16$, we have $f_{\text{patch}} = 32\,Hz$ and $\lfloor S/2 \rfloor = 8$ blind spots:*

m	f_{blind} *(Hz)*	*Interpretation*
1	*32*	*Fundamental patch frequency*
2	*64*	*2nd harmonic*
3	*96*	*3rd harmonic*
4	*128*	*4th harmonic*
5	*160*	*5th harmonic*
6	*192*	*6th harmonic*
7	*224*	*7th harmonic*
8	*256*	*Nyquist limit of token sequence*

Any input signal whose dominant frequency falls on one of these values will produce a degenerate token sequence and trigger a measurable drop in forecasting accuracy.

2.6 Summary

Condition	Effect
$f = m f_s / S$	Full phase-locking → identical patches → constant token sequence.
$f = k f_s / P$	Internal patch periodicity → within-patch redundancy.
Both simultaneously	Maximal degeneracy: constant *and* redundant token sequence.
$S < P$ (overlap)	Partial mitigation; residual phase information between tokens.
$S \geq P$ (no overlap)	Full degeneracy at blind-spot frequencies; no mitigation.

Table 1: *Summary of conditions and their effects on token-space observability.*

3 Point 3

4 Point 4

5 Point 5

6 Point 6

7 Point 7

8 Point 8

9 Point 9

10 Conclusions

References

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